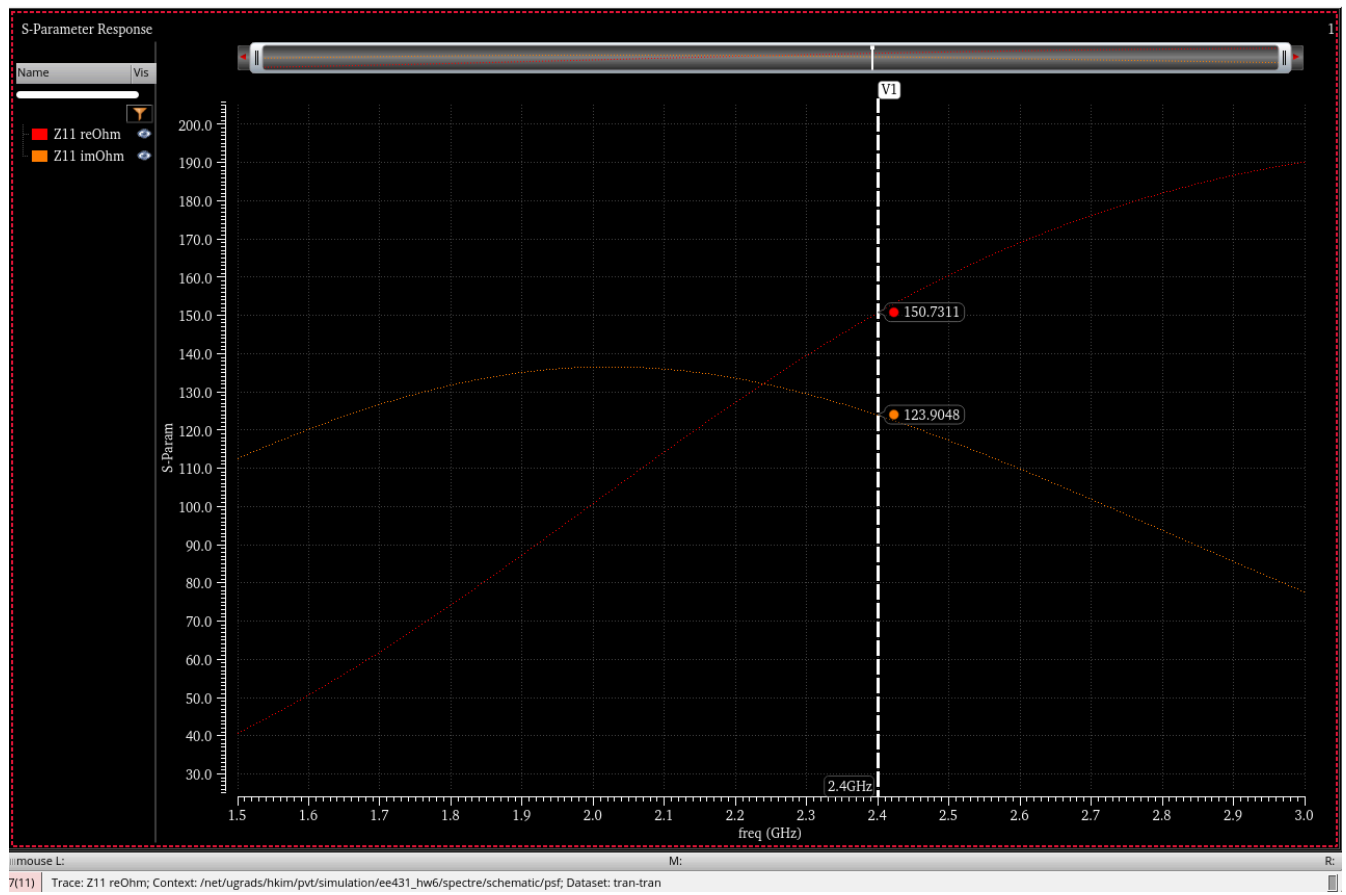
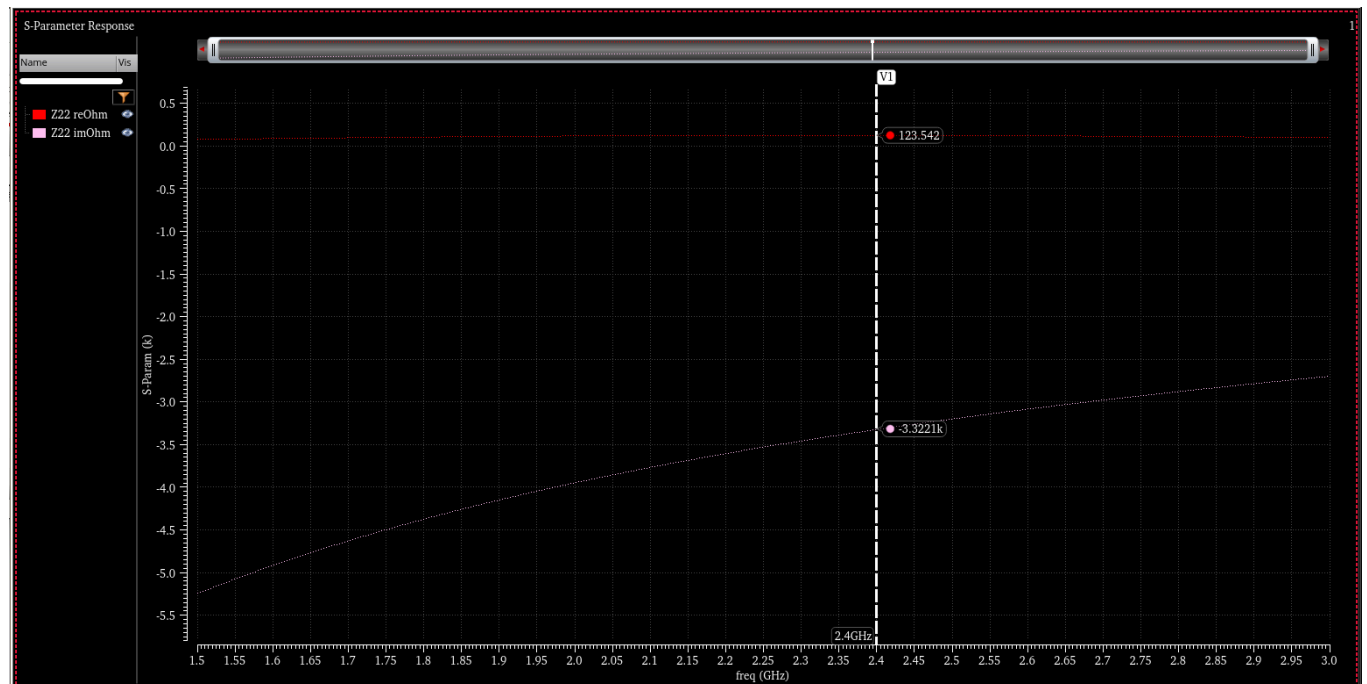


## Input Impedance Plot



S11 Impedance:  $150.7311 + j123.9048$

## Output Impedance Plot



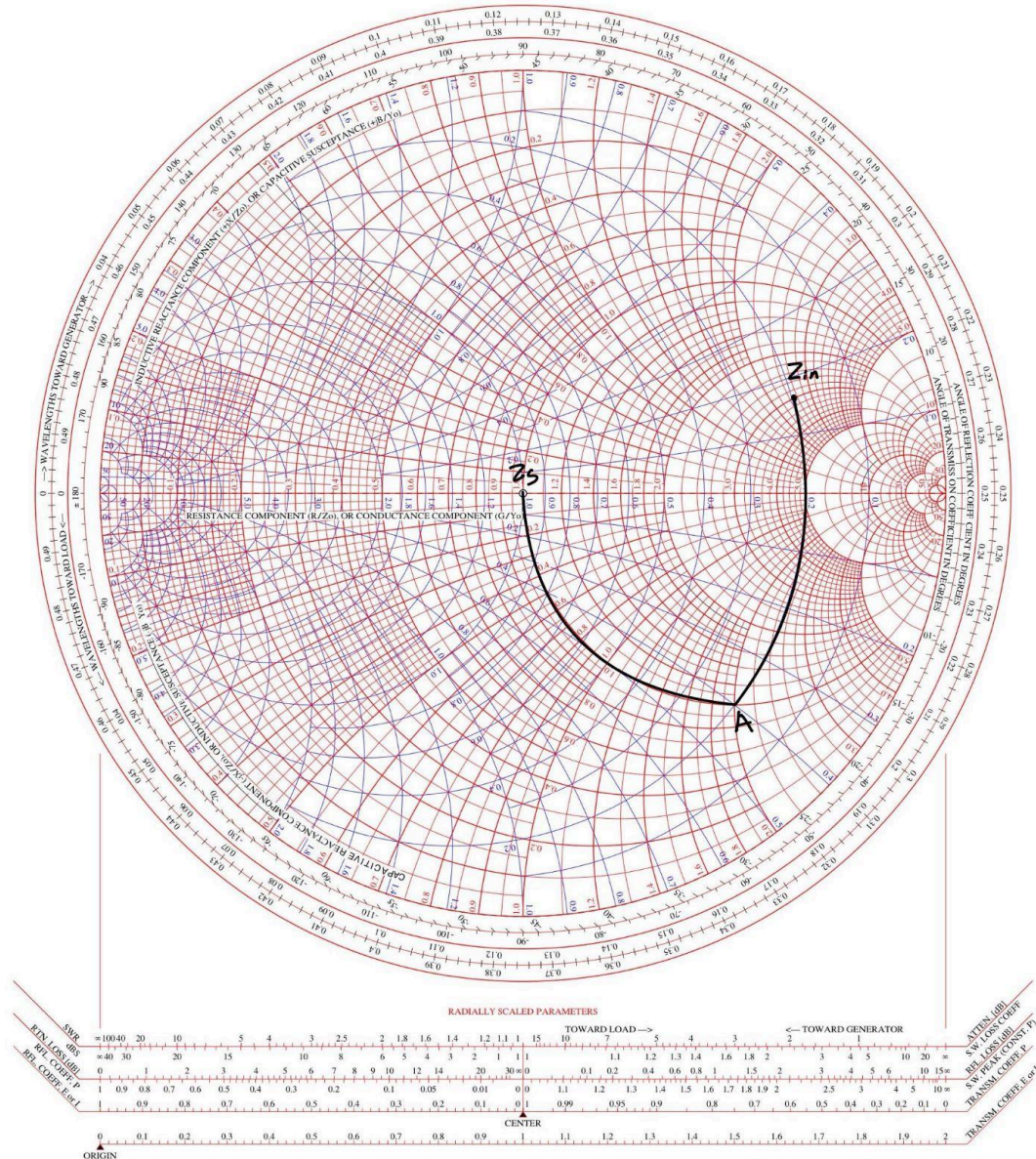
S22 Impedance:  $123.542 - j3.3221k$

# Input Impedance Matching

Smith chart for input matching

|                          |                                              |          |
|--------------------------|----------------------------------------------|----------|
| NAME                     | TITLE<br><i>HW6 Input Impedance</i>          | DWG. NO. |
| SMITH CHART FORM ZY-01-N | Microwave Circuit Design - EE523 - Fall 2000 | DATE     |

## NORMALIZED IMPEDANCE AND ADMITTANCE COORDINATES



Input impedance =  $150.7311 + j123.9048$

normalized impedance =  $150.7311/50 + j123.9048/50 = 3.014 + j2.478$

Capacitance:  $b_{in} = -0.163$ ,  $b_a = 0.4$ ;  $\Delta b = 0.4 + 0.163 = 0.563$

$C = b/(2\pi f Z_o) = 7.47 \cdot 10^{-13} \text{ F} = 0.747 \text{ pF}$

Inductance:  $x_a = -2.0$ ,  $x_s = 0$ ;  $\Delta x = 0 + 2.0 = 2.0$

$L = (x Z_o)/(2\pi f) = 6.63 \cdot 10^{-9} \text{ H} = 6.63 \text{ nH}$

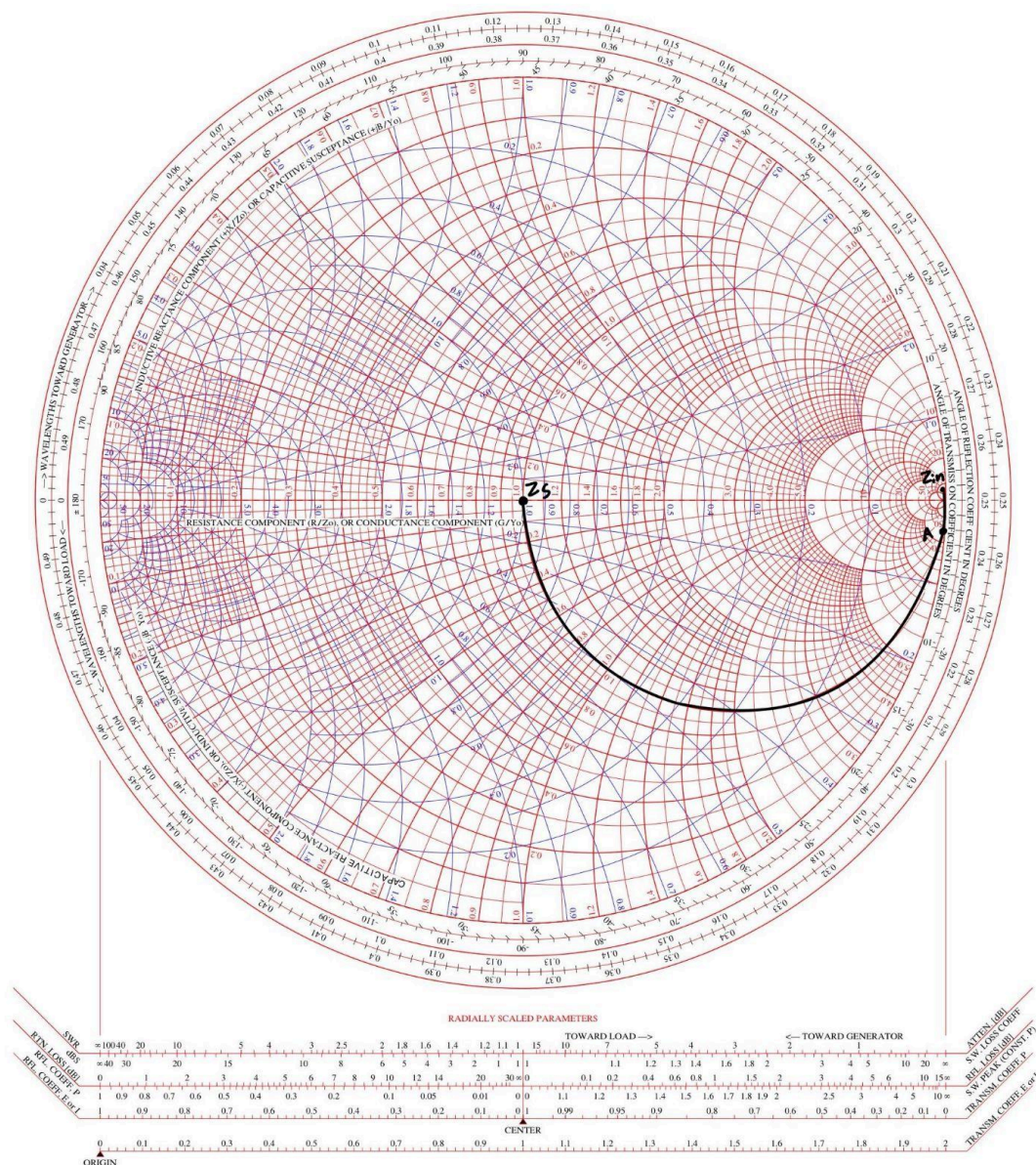
# Output Impedance Matching

Smith chart for input matching

Output im

|                          |                                              |          |
|--------------------------|----------------------------------------------|----------|
| NAME                     | TITLE <b>HW6 Output Impedance</b>            | DWG. NO. |
| SMITH CHART FORM ZY-01-N | Microwave Circuit Design - EE523 - Fall 2000 | DATE     |

## NORMALIZED IMPEDANCE AND ADMITTANCE COORDINATES



pedance =  $123.542 - j3.3221k$

normalized impedance =  $(123.542 - j3.3221k)/50 = 2.471 - j66.442$

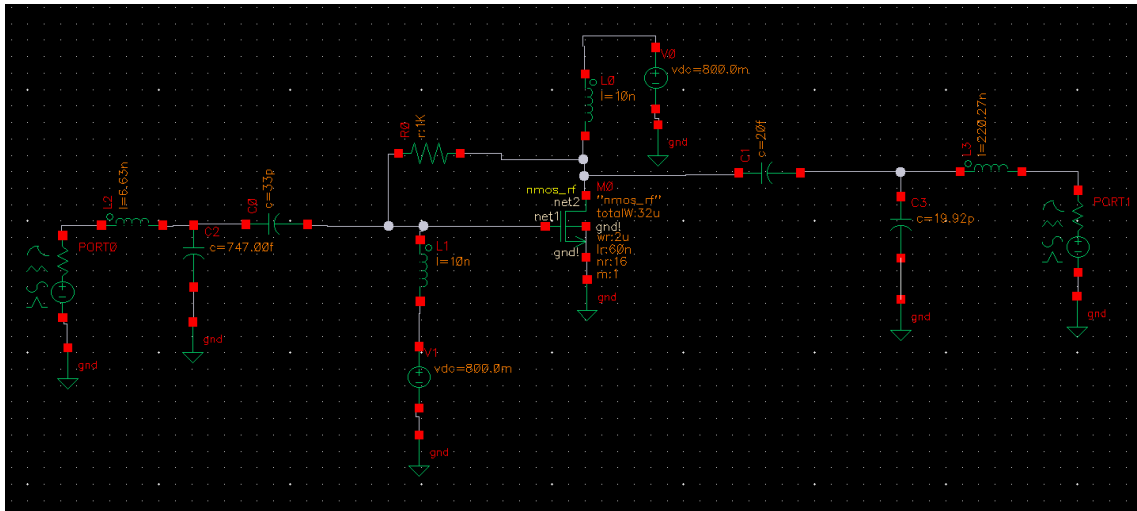
Capacitance:  $b_{in} = 0.015$ ,  $b_a = 15.03$ ;  $\Delta b = 15.03 - 0.015 = 15.02$

$$C = b/(2\pi f Z_o) = 1.992 \times 10^{-11} \text{ F} = 19.92 \text{ pF}$$

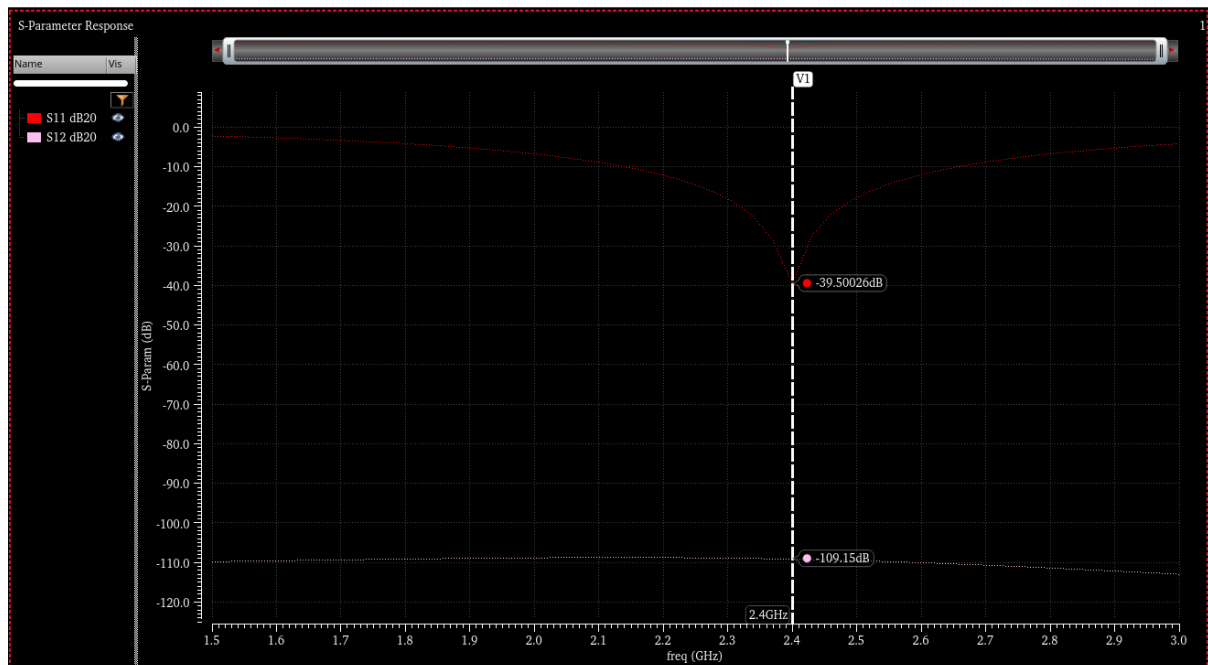
Inductance:  $x_a = -66.44$ ,  $x_s = 0$ ;  $\Delta x = 0 + 2.0 = 66.44$

$$L = (x Z_o)/(2\pi f) = 2.2027 \times 10^{-7} \text{ H} = 220.27 \text{ nH}$$

## Impedance Matching Schematics



## Plot of S11 and S12



## Plot of S22 and S21



## Noise Figure plot

